

2018 H1 Chemistry Prelim Paper 2 Answers

- 1 (a) Amount of NaOH = $35.65/1000 \times 0.018 = \underline{6.42 \times 10^{-4} \text{ mol}}$ [1]

Amount of acid in $250 \text{ cm}^3 = 0.00700 \times 250/25 = \underline{6.42 \times 10^{-3} \text{ mol}}$ [1]

- (b) 1 mol of ascorbic acid reacts with 1 mol of iodine [1]

- (c) Amount of ascorbic acid in 100 ml syrup
 $= [(33.40/1000) \times 0.017] \times (250/25)$
 $= 0.005678$
 $= \underline{5.68 \times 10^{-3} \text{ mol}}$ [1] ecf if b(i) wrong

- (d) Reaction 1 (Neutralisation) – Screened methyl orange/ methyl orange/ thymol blue
 (Reaction 2) Redox reaction – Starch [1]

- (e) Amount of benzoic acid = $0.006417 - 0.005678 = 7.39 \times 10^{-4} \text{ mol}$

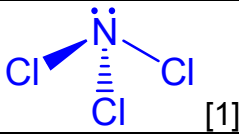
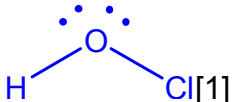
Mass of benzoic acid = $7.39 \times 10^{-4} \times 122.0 = \underline{0.0902 \text{ g}}$ [1]

Mass of ascorbic acid = $0.00568 \times 176.0 = \underline{1.00 \text{ g}}$ [1] ecf in (b)(ii) wrong

- (f) Since 1g (= 1000 mg) of ascorbic acid is found in 100 cm^3 of Appelin Syrup,

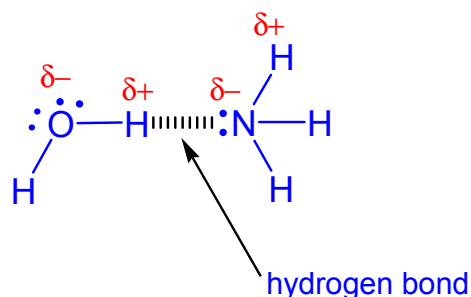
Dosage of Appelin B12 Syrup = $(50/1000) \times 100$
 $= 5 \text{ cm}^3$ [1]

- 2 (a) (i)

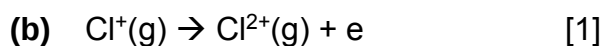
Compound	3D sketch	Shape	Bond Angle
NCl_3		Trigonal pyramidal [1]	107°
HOCl		Bent [1]	105°

[1] for correct bond angles for both molecules

- (ii) The energy released in forming strong hydrogen bonds between ammonia and water molecules is sufficient to overcome the hydrogen bonds between water molecules and the hydrogen bonds between ammonia molecules. [1]



[1] – lone pairs, hydrogen bond, partial charges, label



Cl^+ : $[\text{Ne}]3\text{s}^2 3\text{p}_x^2 3\text{p}_y^1 3\text{p}_z^1$

S^+ : $[\text{Ne}]3\text{s}^2 3\text{p}_x^1 3\text{p}_y^1 3\text{p}_z^1$

Less energy is required to remove the paired 3p_x electron in Cl^+ as it experiences inter-electron repulsion.

[1]

Oxide	Reaction with hydrochloric acid	Reaction with sodium hydroxide
Na_2O	$\text{Na}_2\text{O}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	No reaction
Al_2O_3	$\text{Al}_2\text{O}_3(\text{s}) + 6\text{HCl}(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$	$\text{Al}_2\text{O}_3(\text{s}) + 2\text{NaOH}(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaAl}(\text{OH})_4(\text{aq})$
P_4O_{10}	No reaction	$\text{P}_4\text{O}_{10}(\text{s}) + 12\text{NaOH}(\text{aq}) \rightarrow 4\text{Na}_3\text{PO}_4(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$

[1] for each row

- 3 (a) (i) Isotopes are atoms of the same element which have the same atomic number/number of protons but different nucleon number/number of neutrons. [1]

- (ii) Number of protons : 10
 Number of neutrons: 10
 Number of electrons: 10 [1]

	D_2O	H_2O	HDO
Initial amt/ mol	$30 / 20.0 = 1.5$	$27 / 18.0 = 1.5$	0
Change/ mol	-x	-x	+2x
Eqm amt/ mol	$1.5 - x$	$1.5 - x$	2x

[1] for initial amounts

Let volume of mixture be $V \text{ dm}^3$.

$$\frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{1.5-x}{V}\right)^2} = 3.56$$

$$\frac{2x}{1.5-x} = 1.886 \quad [1] \text{ for correct substitution. Mark for } V?$$

$$x = 0.7279$$

$$n(\text{HDO}) = 2x = 1.46 \text{ mol} [1]$$

*Note if students got M_r of D_2O wrong. They will not be able to solve the wrong quadratic equation (only max 1 out of 3 marks).

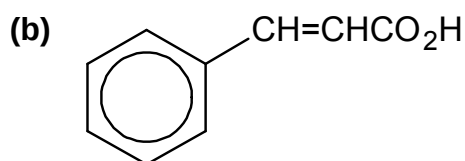
- (c) (i) $[\text{H}^+] = 10^{-6.92} = 1.202 \times 10^{-7} \text{ mol dm}^{-3}$
 Since water is neutral, $[\text{OH}^-] = [\text{H}^+] = 1.202 \times 10^{-7} \text{ mol dm}^{-3}$
 $K_w = [\text{H}^+][\text{OH}^-] = (1.202 \times 10^{-7})^2 = 1.45 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6} [1] \text{ for value}$

- (ii) Value of K_w at 25°C is 1.0×10^{-14}

At higher temperature of 30°C , K_w increases. Positon of equilibrium has shifted to the right and the forward endothermic reaction is favoured. [1]

Hence the forward reaction (auto-ionisation of water) is endothermic.
Enthalpy increases (more positive/ more endothermic) during reaction.
 [1]

- 4 (a) I: Elimination
 II: Oxidation
 III: Condensation
 IV: Acidic hydrolysis

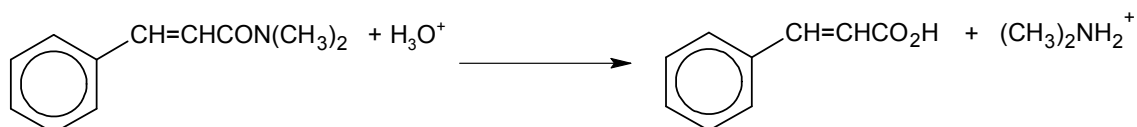


- (c) I: Ethanolic NaOH , heat under reflux – **heat under reflux is a must to be written for this R&C.**

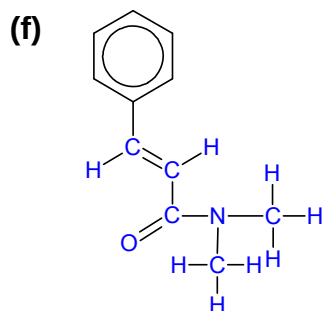
III: $(\text{CH}_3)_2\text{NH}$ or dimethylamine
 N,N' Dicyclohexylcarbodiimide with 5% DMAP catalyst
Many have missed out the amine reagent or included heat under reflux (which was unnecessary).

IV: Dilute H_2SO_4 , heat (under reflux) – **aq or dilute acid should be mentioned but students often missed out.**

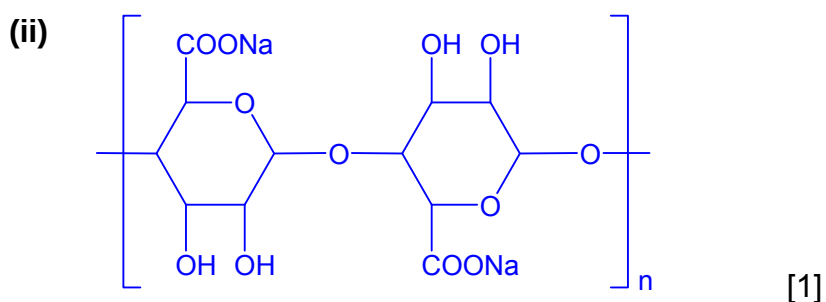
(d) Equation for step IV:



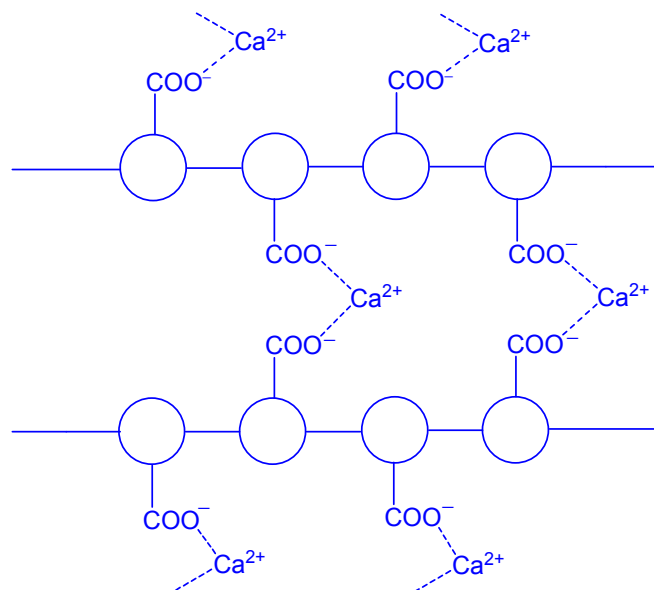
(e) Cis-trans isomerism occurs when there is restricted rotation about a C=C π bond in an aliphatic alkene [1] and when there are two different groups on each carbon of the double bond. [1]



5 (a) (i) Ionic bonds (between oppositely charged ions) and covalent bonds (between atoms) [1]



(b) (i)



[1] – show at least two chains crosslinking

[1] – Ca^{2+} near to two opposite chain COO^- groups

It is held by ionic bonds between one Ca^{2+} and two alginate monomers. [1]

- (ii) The energy released when calcium alginate forms ion-dipole interactions (and hydrogen bonds) with the water molecules is insufficient to overcome the strong ionic bonds between calcium ions and alginate ions. [2]
- (c) (i) Lime juice has a pH lower than 3.6. The high concentration of H^+ in lime juice will acidify sodium alginate to form alginic acid (acid-base reaction) which prevents spherification from occurring. [1]
- (ii) Sardine extract has a very high calcium content and will solidify in the sodium alginate solution immediately to form calcium alginate. [1]
- (d) Mango puree. [1]
Its pH is greater than 3.6 and it has the lowest calcium content. [1]
- (e) $[\text{H}^+]$ from milkfish extract = $10^{-5.4} = 3.981 \times 10^{-6} \text{ mol dm}^{-3}$
 $[\text{H}^+]$ from lime juice = $10^{-1.8} = 0.01585 \text{ mol dm}^{-3}$ [1]

$$[\text{H}^+] \text{ of solution} = \frac{3.981 \times 10^{-6} \left(\frac{50}{1000} \right) + 0.01585 \left(\frac{10}{1000} \right)}{\frac{50 + 10}{1000}} \quad [1]$$

$$= 0.002644 \text{ mol dm}^{-3}$$

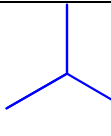
$$\text{pH} = -\log (0.002644) = 2.57 \quad (\text{pH} < 3.6)$$

The solution is too acidic. It is unsuitable for spherification. [1]

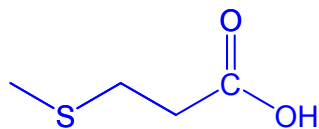
- (f) (i) • There are 3 bond pair and 0 lone pair of electrons about C^* atom.
 • There are 2 bond pairs and 2 bond pairs of electrons about S atom.
 • According to the VSEPR theory, the electron pairs will arrange themselves as far apart as possible to minimise repulsions.
 • Hence the shape about C^* atom is trigonal planar and
 • the shape about S atom is bent.

5 points – [3], 4points – [2], 3-2 points – [1]

(ii)

	Step I	Step II
Type of reaction	Oxidation [1]	Condensation [1]
Reagents and conditions	$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq}), \text{H}_2\text{SO}_4(\text{aq})$, heat under reflux or $\text{KMnO}_4(\text{aq}), \text{H}_2\text{SO}_4(\text{aq})$, heat under reflux [1]	 NH_2 , DCC (N,N'-Dicyclohexylcarbodiimide) with 5% DMAP catalyst [1]

Structure of Compound X:



[1]



Accept $2\text{CO} + \text{NO}_2 \rightarrow \frac{1}{2}\text{N}_2 + 2\text{CO}_2$

(ii) The reactants and catalyst are in different phases. [1]

(iii) Reactant molecules diffuse towards the platinum catalyst surface.

Reactant molecules become chemically adsorbed on neighbouring active sites by forming bonds with the catalyst surface.

This caused the covalent bonds within the molecules to weaken, hence lowering the activation energy of the reaction.

The adsorbed molecules are also held in the right orientation for reaction to occur, hence increasing the frequency of effective "collisions".

Eventually, the molecules dissociate, forming highly reactive intermediates which then react to form the products.

After reaction, the product molecules desorbed and diffuse away from the surface.

[3] for 4 points, [2] for 3 points, [1] for 2 points

(iv) Nanoparticles are discrete particles with all three dimensions in the size range between 1 to 100 nm. [1]

(v) When the particles used are in the nanoscale, they provide a larger surface area to volume ratio compared to particles in larger dimensions.

Hence, there will be greater number of active sites for reactions to occur.

This increases the ratio of platinum ions on the surface of a platinum nanoparticle to platinum ions that are buried inside the same platinum nanoparticle.

[2] for 3 points, [1] for 2 points

(vi) Nanoparticles which are released into the environment might be taken in by bacteria and protozoa through their cell membranes, thus allowing the nanomaterials to enter the biological food chain. [1]

- (b) (i) By Le Chatelier's Principle, a decrease in temperature will favour the exothermic reaction to compensate the lowered heat which is the forward reaction. Hence the position of equilibrium will shift to the right.

The mixture is decolourised.

[2] for 3 points and [1] for 2 points

- (ii) By Le Chatelier's Principle, a decrease in pressure will favour the reaction that produced a greater amount of gas which is the backward reaction. Hence the position of equilibrium shifts to the left.

The brown colour of the mixture is intensified.

[2] for 3 points and [1] for 2 points

- (c) (i) By exerting high pressure [1]
At high pressure, the molecules are pushed closer together, resulting in the forming of intermolecular forces of attraction between the molecules. [1]

or

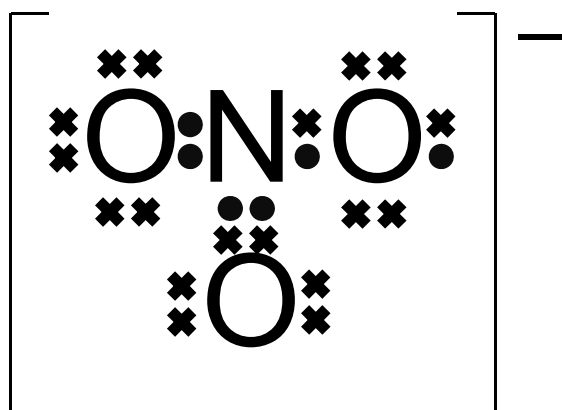
By lowering the temperature [1]
At low temperature, the molecules will move at a slower speed. This helps to bring the molecules closer together, resulting in the forming of intermolecular forces of attraction between the molecules. [1]

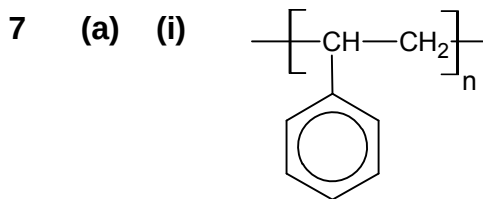
- (ii) $\Delta H_{\text{rxn}} = \sum \text{BE}(\text{bonds broken}) - \sum \text{BE}(\text{bonds formed})$
 $= 160(3) + 390(8) + 201(2) + 607(2) - 944(3) - 460(8)$ [1]
 $= -1296$
 $= -1300 \text{ kJ mol}^{-1}$ [1]

- (iii) Nitrogen tetroxide, hydrazine and water exists as liquids and the enthalpy change of vapourisation of these compounds have to be taken into account. [1]

- (d) (i) $\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$ [1]

- (ii)





[1]

Addition polymerisation [1]

- (ii) Polystyrene is a thermoplastic which has low melting point (or flexible; low tensile strength) while poly (diallyl phthalate) is a thermoset polymer which has high melting point (or rigid; high tensile strength) [1]
- (iii) Polystyrene is highly recyclable or has very good reshaping capability [1]
- (iv) Plastics are finite resources (economic factor)
Harm to environment, human health (environment factor)
[1] for any logical answer

- (b) (i) (By reasoning method)
Comparing experiment 1 and 2
When $[C_9H_9Cl]$ is increased by 1.5 times, the initial rate of reaction increases by approximately $4.9/3.2 \approx 1.5$.
Hence, the order of reaction wrt C_9H_9Cl is 1. [1]

Comparing experiment 2 and 3
When $[C_9H_9Cl]$ is halved and $[NH_3]$ is increased by 4 times, the initial rate of reaction is almost doubled ($6.3/3.2$)
Hence, the order of reaction wrt NH_3 is 1. [1]

Can accept calculation method.

- (ii) Rate = $k[C_9H_9Cl]^1 [NH_3]^1$ [1]

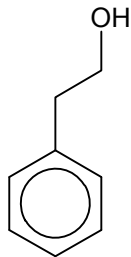
- (iii) Rate = $k[C_9H_9Cl]^1 [NH_3]^1$
 $6.3 \times 10^{-3} = k (0.01)(0.4)$
 $k = 1.58 [1] \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1] including units

- (c) (i) Rate of hydrolysis depends on the strength of the C–X bond.
- Down the group, bond length increases: C–Cl < C–Br < C–I
 - Orbital overlap becomes less effective down the group.
 - Strength of C–X bond: C–Cl (340) > C–Br (280) > C–I (240)
Or atomic radius: Cl (0.099) < Br (0.114) < I (0.133)
 - Rate of hydrolysis: $C_9H_9Cl < C_9H_9Br < C_9H_9I$
Amount of ppt from R–X: $C_9H_9Cl < C_9H_9Br < C_9H_9I$
- [2] for 4 points

(ii) Pale cream precipitate [1]

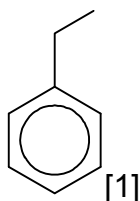
- (d) (i) Compound **B** [1]
Primary alcohol can be oxidized to form carboxylic acid. Since **D** is a carboxylic acid, **B** has to be a primary alcohol. [1]

(ii)



Primary alcohol: [1]

(iii)



(iv) Compound **F** is an ester. [1]

(v) Add Br₂ in CCl₄ to separate test tubes of styrene and C. [1]

Observation

Styrene: Orange-red Br₂ decolourises

C: Orange-red Br₂ remains [1]

OR

Add KMnO₄, dilute H₂SO₄ to separate test tubes of styrene and C

Observation

Styrene: Purple colour decolourises

C: Purple colour remains